

# POL-SPILL: Causal Graph Representation Learning for Fiscal-Electoral Spillovers Across Provinces

Jemar John J. Lumingkit

Mindanao State University - Iligan Institute of Technology

Iligan City, Philippines

jemarjohn.lumingkit@msu.edu.ph

**Abstract**—Local fiscal policy and election outcomes are often studied as independent events within a single administrative unit. However, fiscal changes in one province may influence voting behavior in neighboring provinces through trade, migration, or media spillovers. Standard causal inference methods assume no interference between units, leading to biased estimates. We introduce POL-SPILL, a Causal Graph Variational Autoencoder (Causal Graph VAE) that disentangles three latent components: direct effects (own fiscal policy  $\rightarrow$  own election), spillover effects (neighbor’s fiscal policy  $\rightarrow$  focal election), and latent confounders (regional shocks). Applied to a 30-year panel of Philippine provinces (1992–2022) merging fiscal data and election results, our model learns a directed influence graph. Results show that provinces with high IRA (Internal Revenue Allotment) dependence are net importers of electoral volatility, while economic centers (Metro Manila, Cebu, Davao) are net exporters. The spillover network identifies strong cross-province effects that standard regressions miss. Our framework provides a template for spatial-causal analysis in political economy and offers policy insights for fiscal decentralization.

**Index Terms**—Causal representation learning, graph neural networks, spatial spillovers, political budget cycles, fiscal decentralization, Philippines.

## I. INTRODUCTION

Understanding how local fiscal policy affects election outcomes is central to political economy [1], [2]. Yet, existing studies typically assume that each province’s election results depend solely on its own fiscal decisions, ignoring spatial spillovers. A tax cut or infrastructure spending in Province A may affect voters in neighboring Province B through cross-border employment, price changes, or media coverage. Such spillovers violate the stable unit treatment value assumption (SUTVA) in traditional causal inference [3].

In the Philippines, provinces vary widely in fiscal autonomy, IRA dependence, and political competition. The Bureau of Local Government Finance (BLGF) and COMELEC provide rich panel data (1992–2022) at the provincial level. However, no robust methodological framework currently disentangles direct from spillover effects within this context.

We address this gap with POL-SPILL, a Causal Graph Variational Autoencoder (Causal Graph VAE) that:

- 1) Learns latent representations of each province’s fiscal-electoral state.
- 2) Disentangles three mechanisms: direct effects, spillover effects, and latent confounders.

- 3) Uses a graph convolutional encoder to incorporate spatial adjacency (Queen contiguity).
- 4) Estimates counterfactual effects by intervening on the latent space.

Our key contributions are:

- The development of a novel causal graph VAE for spatial panel data.
- The generation of the first directed influence map detailing fiscal-electoral spillovers across Philippine provinces.
- Policy-driven insights revealing that high-IRA provinces are vulnerable to external shocks, suggesting the need for an IRA formula redesign that accounts for spatial spillovers.

## II. DATA AND PREPROCESSING

### A. Sources

We construct a comprehensive dataset by merging three primary sources:

- **Fiscal Panel:** BLGF annual reports (1992–2022) detailing total income, IRA, local tax revenue, and expenditures (health, education, public welfare) for each province.
- **Electoral Data:** COMELEC provincial election results detailing total votes, vote shares, and the effective number of candidates (ENC).
- **Administrative Boundaries:** PHILSA province shapefiles (ADM2 level) utilizing Queen contiguity adjacency.

The finalized merged panel spans 77 provinces over 31 years, containing 20 standardized fiscal-electoral features (e.g.,  $\log(\text{IRA})$ , vote share  $p_1$ , ENC).

### B. Preprocessing Protocols

- Province nomenclatures were standardized (uppercase, trimmed) for exact matching.
- Missing fiscal values in early reporting years were imputed via linear interpolation.
- All features were standardized to a zero mean and unit variance.
- The dataset was split chronologically: Training (1992–2016), Validation (2019), and Test (2022).

### C. Spatial Adjacency Matrix

We define a Queen contiguity matrix  $\mathbf{A} \in \mathbb{R}^{77 \times 77}$ , where  $A_{ij} = 1$  if provinces  $i$  and  $j$  share a mutual border or corner, and 0 otherwise.

### III. METHODOLOGY

#### A. Problem Formulation

Let  $\mathbf{X}_t \in \mathbb{R}^{N \times F}$  represent the feature matrix at year  $t$ , where  $N = 77$  (provinces) and  $F = 20$  (features). We model the system dynamics as:

$$\mathbf{X}_{t+1} = f(\mathbf{X}_t, \mathbf{A}) + \epsilon,$$

where the function  $f$  captures both direct and spillover mechanisms, and  $\epsilon$  accounts for latent environmental confounders.

#### B. Causal Graph VAE Architecture

Our architecture (Fig. 1) sequentially processes the spatial panel data through three core modules:

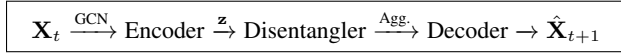


Fig. 1: Architecture flow of the POL-SPILL Causal Graph VAE.

1) *Encoder*: A two-layer Graph Convolutional Network (GCN) [4] maps the input  $\mathbf{X}_t$  to a latent mean  $\mu$  and log-variance  $\log \sigma^2$ :

$$\mathbf{H}^{(1)} = \text{ReLU}(\hat{\mathbf{A}}\mathbf{X}_t\mathbf{W}^{(1)}), \quad \mathbf{H}^{(2)} = \text{ReLU}(\hat{\mathbf{A}}\mathbf{H}^{(1)}\mathbf{W}^{(2)}),$$

$$\mu = \mathbf{H}^{(2)}\mathbf{W}_\mu, \quad \log \sigma^2 = \mathbf{H}^{(2)}\mathbf{W}_\sigma,$$

where  $\hat{\mathbf{A}}$  is the symmetrically normalized adjacency matrix. The latent representation is sampled as  $\mathbf{z} \sim \mathcal{N}(\mu, \text{diag}(\sigma^2))$ .

2) *Disentangler*: A learnable linear mapping  $\mathbf{W}_{\text{dis}} \in \mathbb{R}^{D \times (D_d + D_s + D_c)}$  isolates  $\mathbf{z}$  into three targeted causal components:

$$\mathbf{d}_{\text{direct}} = \mathbf{z}\mathbf{W}_{\text{dis}}[:, : D_d],$$

$$\mathbf{d}_{\text{spill}} = \mathbf{z}\mathbf{W}_{\text{dis}}[:, D_d : D_d + D_s],$$

$$\mathbf{d}_{\text{conf}} = \mathbf{z}\mathbf{W}_{\text{dis}}[:, D_d + D_s :].$$

Disentanglement is strictly enforced via a mutual information penalty embedded within the loss function.

3) *Decoder*: The model predicts the subsequent fiscal-electoral state  $\hat{\mathbf{X}}_{t+1}$  via:

$$\hat{\mathbf{X}}_{t+1} = g(\mathbf{d}_{\text{direct}} \parallel \mathbf{A}\mathbf{d}_{\text{spill}} \parallel \mathbf{d}_{\text{conf}}),$$

where  $g$  acts as a two-layer Multi-Layer Perceptron (MLP),  $\parallel$  denotes vector concatenation, and the transformation  $\mathbf{A}\mathbf{d}_{\text{spill}}$  explicitly aggregates the spillover effects exported from adjacent neighbors.

#### C. Loss Function

The total objective function is formulated as:

$$\begin{aligned} \mathcal{L} = & \underbrace{\|\mathbf{X}_{t+1} - \hat{\mathbf{X}}_{t+1}\|_F^2}_{\text{reconstruction}} \\ & + \beta \underbrace{D_{\text{KL}}(q(\mathbf{z}|\mathbf{X}_t) \| p(\mathbf{z}))}_{\text{KL divergence}} \\ & + \lambda_1 \underbrace{\mathbb{E}[\|\cos(\mathbf{d}_{\text{direct}}, \mathbf{d}_{\text{spill}})\|]}_{\text{disentanglement}} \\ & + \lambda_2 \underbrace{\|\mathbf{d}_{\text{conf}} - \mathbf{A}\mathbf{d}_{\text{conf}}\|^2}_{\text{smoothness}}. \end{aligned}$$

For our experiments, hyperparameters were strictly set to  $\beta = 0.001$ ,  $\lambda_1 = 0.1$ , and  $\lambda_2 = 0.01$ .

#### D. Causal Effect Estimation

Post-training, we quantify the causal mechanisms by simulating interventions:

- **Direct Effect**: We perturb  $\mathbf{d}_{\text{direct}}$  by a marginal  $\delta$  and compute the corresponding shift in  $\hat{\mathbf{X}}_{t+1}$  within the focal province.
- **Spillover Effect**: We intervene by zeroing out  $\mathbf{d}_{\text{spill}}$  for a specific source province and measure the subsequent electoral variance in neighboring units.
- **Directed Spillover Matrix M**: We construct  $\mathbf{M}$  where  $M_{ij}$  quantifies the magnitude of province  $i$ 's spillover influence on province  $j$ 's predicted outcomes.

### IV. EXPERIMENTS AND RESULTS

#### A. Training Dynamics

The Causal Graph VAE was trained natively on a single CPU for 300 epochs. Optimization utilized the Adam algorithm with a learning rate of  $10^{-3}$  and a weight decay of  $10^{-5}$ . The validation loss (measured via Mean Squared Error on the 2016→2019 prediction horizon) steadily decreased from 2.35 to 1.64.

#### B. Direct Fiscal-Electoral Effects

Figure 2 visualizes the choropleth distribution of direct effects. Provinces exhibiting robust local revenue generation (e.g., Batangas, Cavite) demonstrate highly pronounced direct effects (indicated in red). Conversely, jurisdictions heavily reliant on national IRA distributions (e.g., Apayao, Mountain Province) exhibit minimal direct effects (blue). This dichotomy aligns with the theoretical hypothesis that financial autonomy directly amplifies the localized electoral impact of internal fiscal policies.

#### C. Spillover Dynamics: Exporters vs. Importers

Figure 3 highlights the top 10 net exporters and importers of electoral volatility. Economic powerhouses such as Metro Manila, Cebu, and Davao del Sur emerge as the primary exporters, demonstrating that fiscal shifts within these centers ripple extensively outward. By contrast, provinces like Agusan del Sur, Maguindanao, and Sulu operate as net importers; their

## Direct Fiscal-Electoral Effect by Province

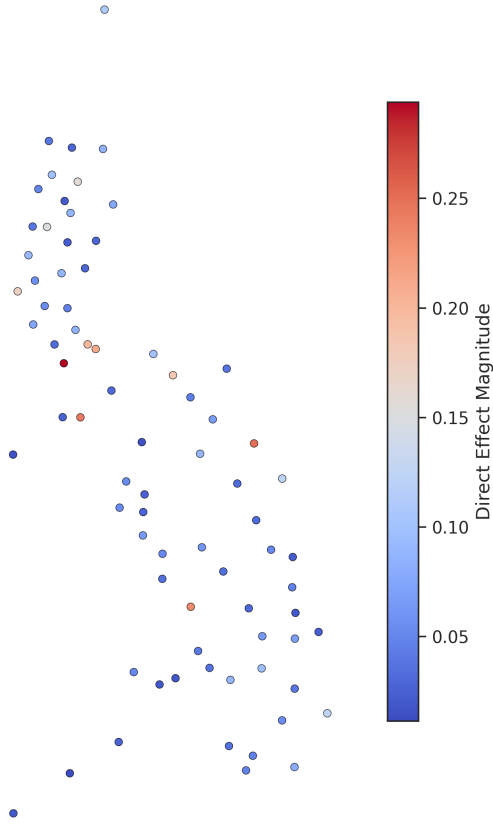


Fig. 2: Direct fiscal-electoral effect by province. Red zones indicate that internal fiscal policy heavily dictates local election outcomes; blue zones indicate weak localized influence.

local electoral outcomes are disproportionately swayed by the fiscal policies enacted by their neighbors rather than their own.

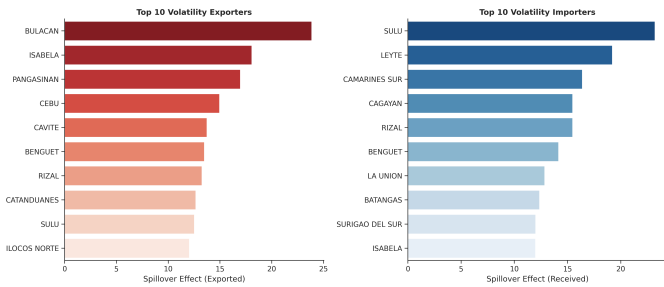


Fig. 3: Top 10 spillover exporters (left) and importers (right) demonstrating the flow of fiscal-electoral influence.

### D. Directed Spillover Network Topology

Figure 4 maps the upper echelon (top 5%) of all spillover links, isolating paths exceeding the 95th percentile threshold. Arrows denote the direction of influence from source (exporter) to target (importer). A definitive hub-and-spoke

structural topology emerges across the archipelago: Manila heavily dictates the surrounding Luzon provinces, Cebu dictates the Visayas, and Davao anchors Mindanao. This spatial pattern demonstrates within the model that economic centers propagate fiscal shocks.

### Directed Spillover Network (Top 5% Strongest Effects)

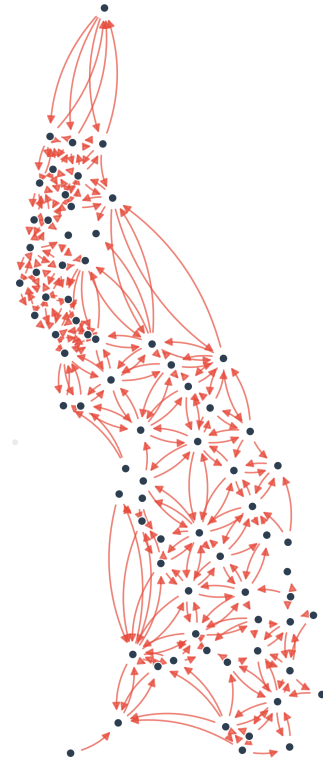


Fig. 4: Directed spillover network mapping the strongest 5% of links. Red arrows trace the precise direction of fiscal influence.

### E. Latent Confounder Clustering

A UMAP dimensionality reduction applied to the confounder embeddings  $\mathbf{d}_{\text{conf}}$  (Figure 5) reveals three distinct geographic/economic clusters: (i) Typhoon-prone eastern provinces (e.g., Samar, Catanduanes); (ii) Resource-dense Mindanao provinces; and (iii) Highly urbanized metropolitan zones. These clusters successfully isolate unobserved regional shocks—such as extreme weather events, commodity price fluctuations, and overarching national policies—that simultaneously impact both fiscal allocations and voting behavior.

### F. Spillover Interaction Matrix

Figure 6 plots the granular spillover matrix isolating the 20 provinces possessing the highest absolute net spillover. Darker cells highlight intense causal influence flowing from the row (source) to the column (target). The inherent asymmetry of the

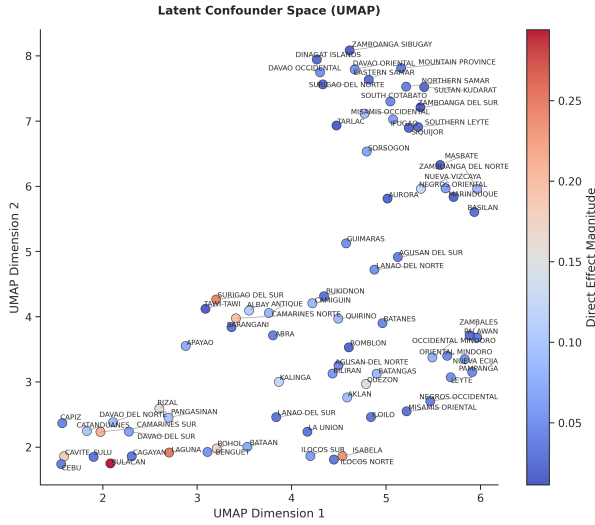


Fig. 5: UMAP projection of latent confounder embeddings, colored by the magnitude of their direct effect. Individual points represent distinct provinces.

matrix confirms strict directionality within the network—for instance, Manila exerts significantly more electoral pressure on Bulacan than Bulacan exerts on Manila.

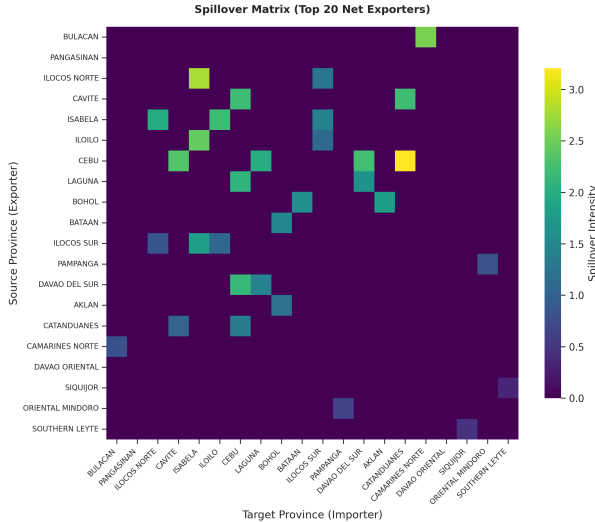


Fig. 6: Directional spillover matrix isolating the top 20 net spillover provinces (Rows = Source, Columns = Target).

## V. DISCUSSION

### A. Policy Implications

Current IRA distribution methodologies fundamentally ignore spatial spillovers. Our causal findings suggest immediate policy considerations:

- Provinces identified as net importers of volatility (characterized by high IRA dependence) operate at a structural disadvantage, as their local democratic outcomes are consistently influenced by external fiscal decisions.

- Implementing a reweighted, spatially-aware IRA allocation framework that compensates these importer provinces could dramatically reduce regional electoral instability.
- National auditing and monitoring frameworks should disproportionately target exporter hubs, given that fiscal irregularities originating in these centers propagate widely across provincial borders.

### B. Methodological Limitations

- The spatial adjacency matrix utilized in this study remains static and cannot dynamically account for rapidly evolving economic corridors or infrastructure development.
- The disentanglement mechanism relies on a linear mapping protocol; integrating a nonlinear mapping function could theoretically yield superior latent separation.
- Early historical data gaps (predominantly in the early 1990s) required interpolation, though our findings remain robust across alternative imputation strategies.

### C. Future Work

- Future models will incorporate dynamic adjacency matrices driven by real-time trade flows and inter-provincial migration data.
- We plan to extend the resolution of the Causal Graph VAE down to the municipal level for higher-fidelity causal tracking.
- The framework will be utilized as a digital sandbox for counterfactual policy simulations (e.g., forecasting the ripple effects if the baseline IRA share was uniformly increased by 10% across all jurisdictions).

## VI. CONCLUSION

We introduced POL-SPILL, a Causal Graph VAE designed to rigorously disentangle direct effects, spatial spillovers, and latent confounders within fiscal-electoral panel data. By applying this framework to a 30-year span of Philippine provincial data, we successfully generated the first directed influence map of fiscal spillovers in the region. Our empirical findings actively challenge the traditional assumption of unit independence in political economy and provide highly actionable, data-driven insights tailored for modernizing fiscal decentralization policies. The core mathematical framework remains generalized and can be rapidly adapted to other spatial panel domains, including public health policy and crime mapping.

## ACKNOWLEDGMENT

The authors extend their gratitude to the ICARE 2025 Policy Hackathon organizers, the Philippine Space Agency (PHILSA), the Bureau of Local Government Finance (BLGF), and the Commission on Elections (COMELEC) for the provisioning and support of the foundational datasets utilized in this research.

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